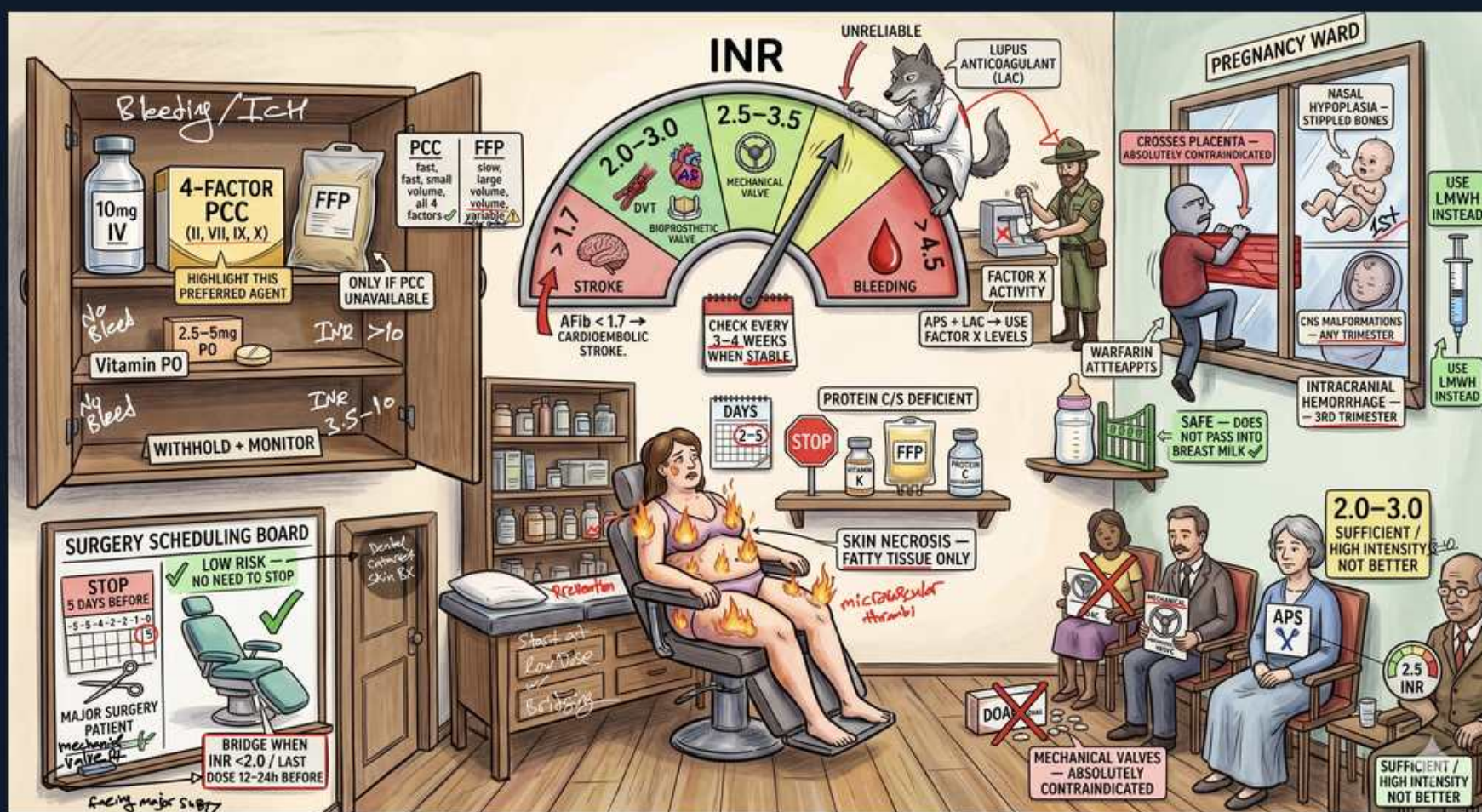


# Warfarin — INR Targets & Reversal





## 1. Target 2.0-3.0

### WHAT YOU SEE

Green band on the big INR speedometer reading '2.0-3.0' with little AFib heart and DVT/clot icons sitting in it — the safe green zone for the common indications.

### WHAT IT ENCODES

The standard INR target is 2.0-3.0 (goal 2.5) for the vast majority of indications: atrial fibrillation, treatment/prevention of VTE (DVT/PE), bioprosthetic valves (first 3 months), and most other indications. In nonvalvular AFib, anticoagulate when CHA2DS2-VASc  $\geq 2$  in men or  $\geq 3$  in women. INR  $< 2.0$  is subtherapeutic (thrombosis risk), INR  $> 3.0$  raises bleeding.

### BOARD TRAP

WRONG: 'push the INR to 3.5-4 in AFib for stronger stroke protection.' Above an INR of  $\sim 3$  the bleeding risk (especially intracranial hemorrhage) rises sharply with no added thromboembolic benefit — 2-3 is the AFib target, not higher.

## 2. Target 2.5-3.5

### WHAT YOU SEE

Amber/higher band on the INR gauge reading '2.5-3.5' beside a mechanical-valve icon — pushed further right (higher target) because the mechanical valve is more thrombogenic.

### WHAT IT ENCODES

A HIGHER target of 2.5-3.5 (goal 3.0) is used for higher-thrombotic-risk mechanical valves — specifically mechanical MITRAL valves, and mechanical aortic valves with additional risk factors (AFib, prior thromboembolism, LV dysfunction). Low-risk newer-generation mechanical aortic valves target 2.0-3.0. Low-dose aspirin is often added.

### BOARD TRAP

WRONG: 'a mechanical mitral valve can be managed with an INR target of 2.0-3.0 like AFib.' The mitral position is highly thrombogenic and requires the higher 2.5-3.5 target — under-targeting risks valve thrombosis and stroke.

## 3. Mechanical valve = warfarin

### WHAT YOU SEE

Seated patient holding a mechanical-valve sign stamped with a red X over 'DOAC' — the red X over DOACs means only warfarin is allowed for mechanical valves.

### WHAT IT ENCODES

Mechanical heart valves REQUIRE warfarin; DOACs are contraindicated. The RE-ALIGN trial showed dabigatran caused MORE thromboembolism AND more bleeding than warfarin in mechanical valves, because contact activation on the artificial surface drives clotting that direct thrombin/Xa inhibitors fail to control. Warfarin is also preferred in moderate-to-severe (rheumatic) mitral stenosis.

### BOARD TRAP

WRONG: 'switch a mechanical-valve patient from warfarin to apixaban for convenience.' DOACs are absolutely contraindicated in mechanical valves (RE-ALIGN: worse outcomes) — warfarin is the only approved oral anticoagulant for them.

## 4. INR 2.5-10, no bleed

### WHAT YOU SEE

Cabinet shelf note 'INR  $> 10 \rightarrow 2.5-5$  mg PO' and 'INR 3.5-10  $\rightarrow$  withhold + monitor' tagged 'No bleed' — no-bleeding INRs get holding  $\pm$  low-dose oral vitamin K, not reversal products.

### WHAT IT ENCODES

Supratherapeutic INR WITHOUT significant bleeding is managed by withholding warfarin and rechecking. INR 4.5-10 with no bleeding: simply hold warfarin (routine vitamin K is NOT recommended). INR  $> 10$  with no bleeding: hold warfarin AND give low-dose ORAL vitamin K 2.5-5 mg. INR slightly above range (3-4.5): lower or hold a dose. Always look for the cause (drug interaction, diet, illness).

### BOARD TRAP

WRONG: 'give FFP or 4-factor PCC for an INR of 8 in a patient who is not bleeding.' Reversal products (PCC/FFP) and high-dose IV vitamin K are reserved for serious bleeding — for asymptomatic high INR you hold warfarin  $\pm$  low-dose oral vitamin K, avoiding overcorrection and rebound thrombosis.

## 5. 4-factor PCC

### WHAT YOU SEE

4-FACTOR PCC (II,VII,IX,X) 10mg IV — preferred

### WHAT IT ENCODES

For MAJOR or life-threatening bleeding on warfarin, 4-factor prothrombin complex concentrate (4F-PCC; contains factors II, VII, IX, X plus proteins C and S) is the PREFERRED reversal agent — dosed by weight and INR, it works within minutes, requires small volume, and needs no blood-type matching. ALWAYS co-administer IV vitamin K because PCC's effect is short-lived.

### BOARD TRAP

WRONG: 'give 4F-PCC alone and skip vitamin K because the INR corrected immediately.' PCC's transfused factors are consumed within hours and the INR will rebound; IV vitamin K must be given concurrently to sustain reversal by restoring endogenous factor synthesis.

## 6. IV vitamin K

### WHAT YOU SEE

Vitamin K IV alongside

### WHAT IT ENCODES

In major bleeding give IV vitamin K 5-10 mg (slow infusion) together with PCC/FFP — it provides DURABLE reversal by allowing hepatic synthesis of functional factors over  $\sim 12-24$  h, covering the gap after PCC wanes. IV (not oral) is used for serious bleeding because it acts faster. Give slowly due to a small risk of anaphylactoid reaction.

### BOARD TRAP

WRONG: 'give a large IV vitamin K dose (e.g. 10 mg) to a stable patient just to normalize the INR fast.' Over-reversing with high-dose vitamin K in a non-bleeding patient causes prolonged warfarin RESISTANCE for days, making re-anticoagulation difficult — high-dose IV vitamin K is for active major bleeding only.



## 7. FFP only if no PCC

### WHAT YOU SEE

Plasma bag labelled 'FFP — ONLY IF PCC UNAVAILABLE / slow, large volume' on the bleeding shelf — flagged as the fallback, second to PCC.

### WHAT IT ENCODES

Fresh frozen plasma is SECOND-LINE for warfarin reversal, used only when 4F-PCC is unavailable. FFP requires large volumes (risk of volume overload/TACO), takes time to thaw and type-match, and reverses less completely/quickly than PCC. It supplies all coagulation factors but is inferior for urgent warfarin reversal.

### BOARD TRAP

WRONG: 'reach for FFP first in a warfarin patient with intracranial hemorrhage.' 4F-PCC is preferred (faster, smaller volume, more complete); FFP is only a fallback when PCC is not on hand — delaying with FFP in ICH costs critical time.

## 8. Stop 5 days pre-op

### WHAT YOU SEE

Surgery scheduling board reading 'STOP 5 DAYS BEFORE' with a 5-day countdown strip — five days off to let the INR drift below 1.5.

### WHAT IT ENCODES

Hold warfarin about 5 days before major surgery to let the INR drift down to <1.5 (≈5 days = ~5 half-lives covering the long-lived factor II). Check INR the day before surgery; if still ≥1.5 give low-dose oral vitamin K 1-2.5 mg. Resume warfarin 12-24 h post-op once hemostasis is secure.

### BOARD TRAP

WRONG: 'stop warfarin the night before surgery.' Warfarin's effect (via long-half-life prothrombin) persists for days; a single night is not enough — you need ~5 days for the INR to normalize before major surgery.

## 9. Bridge if high risk

### WHAT YOU SEE

Scheduling-board note 'BRIDGE when INR <2.0 / last dose 12-24h before' — start LMWH only once warfarin's INR drops below 2, reserved for high thrombotic risk.

### WHAT IT ENCODES

Peri-procedural BRIDGING with therapeutic LMWH (or UFH) is reserved for HIGH thrombotic risk: mechanical mitral valve, recent (<3 mo) VTE/stroke, CHA2DS2-VASc ≥7-9, or known caged-ball valves. Start LMWH when INR falls <2, stop LMWH ~24 h pre-op, resume after hemostasis. The BRIDGE trial showed most AFib patients do NOT benefit from bridging.

### BOARD TRAP

WRONG: 'bridge every warfarin patient with LMWH around surgery.' The BRIDGE trial found that in low/moderate-risk AFib, bridging did not reduce thromboembolism but TRIPLED major bleeding — only high-thrombotic-risk patients should be bridged.

## 10. Skin necrosis

### WHAT YOU SEE

Woman in the chair with purple flame-like necrotic patches over breast/thigh/abdomen, captioned 'SKIN NECROSIS — FATTY TISSUE ONLY' — necrosis literally striking the fatty areas from early protein-C drop microthrombi.

### WHAT IT ENCODES

Warfarin-induced skin necrosis appears within the first ~3-5 days, striking fatty areas (breast, thigh, buttock, abdomen) as painful purple lesions progressing to necrosis. It results from the early steep fall of protein C (short half-life) before procoagulant factors decline, causing microvascular thrombosis — risk is highest in protein C or protein S deficiency.

### BOARD TRAP

WRONG: 'warfarin skin necrosis means the patient is over-anticoagulated, so stop warfarin and wait.' It is an EARLY hypercoagulable complication, not over-anticoagulation; management is stop warfarin, give vitamin K, start a non-warfarin anticoagulant (heparin), and replace protein C — and always bridge with heparin to prevent it.

## 11. Crosses placenta

### WHAT YOU SEE

Pregnancy-ward panel of a fetus behind a red 'CROSSES PLACENTA — ABSOLUTELY CONTRAINDICATED' banner with stippled bones/nasal hypoplasia — warfarin physically crossing the placenta to harm the fetus.

### WHAT IT ENCODES

Warfarin crosses the placenta and is TERATOGENIC: first-trimester exposure causes fetal warfarin syndrome (nasal/bone hypoplasia, stippled epiphyses); later exposure causes CNS abnormalities and fetal/intracranial hemorrhage. It is contraindicated in pregnancy (the narrow exception is some mechanical-valve patients in specialized centers, with counseling).

### BOARD TRAP

WRONG: 'continue warfarin through pregnancy in a VTE patient because heparin is inconvenient.' Warfarin's teratogenicity and fetal bleeding risk mandate switching to LMWH; warfarin should not be used in pregnancy for routine VTE.

## 12. Use LMWH in pregnancy

### WHAT YOU SEE

Green 'USE LMWH INSTEAD' syringe callout in the pregnancy ward — the safe substitute that, being large, cannot cross the placenta.

### WHAT IT ENCODES

LMWH is the anticoagulant of choice in pregnancy because it does NOT cross the placenta (large molecule) — it is not teratogenic and does not cause fetal bleeding. It requires dose adjustment as weight/GFR change and may need anti-Xa monitoring. Unfractionated heparin is an alternative, also placenta-safe.

### BOARD TRAP

WRONG: 'use a DOAC in pregnancy as a convenient warfarin alternative.' DOACs are NOT recommended in pregnancy (insufficient safety data, cross the placenta); LMWH is the correct substitute for warfarin in pregnant patients.

## 13. Safe in breastfeeding

### WHAT YOU SEE

Baby bottles labelled 'SAFE — DOES NOT PASS INTO BREAST MILK' — warfarin is protein-bound and stays out of milk, so lactation is fine.

### WHAT IT ENCODES

Warfarin is COMPATIBLE with breastfeeding — it is highly protein-bound and does not pass into breast milk in clinically significant amounts, so it does not anticoagulate the infant. This contrasts sharply with its pregnancy contraindication; the placenta and breast milk are different barriers.

### BOARD TRAP

WRONG: 'a breastfeeding mother needing anticoagulation must avoid warfarin like she did in pregnancy.' Warfarin (and LMWH/UFH) are all safe during lactation — the teratogenic/placental concern does not apply to breast milk.



## 14. LAC = unreliable INR

### WHAT YOU SEE

Lupus-anticoagulant wolf above the INR gauge tagged 'UNRELIABLE' — the LAC antibody sabotages the INR reading, so the dial can't be trusted.

### WHAT IT ENCODES

In antiphospholipid syndrome with a lupus anticoagulant (LAC), the antibody can interfere with the phospholipid-dependent PT/INR assay, making the INR unreliable for monitoring warfarin. Use chromogenic factor X activity (target ~20-40%) to gauge true anticoagulation intensity. APS VTE is treated with warfarin (DOACs inferior in triple-positive APS).

### BOARD TRAP

WRONG: 'trust the standard INR to dose-adjust warfarin in a lupus-anticoagulant patient.' The LAC artifactually alters the INR, so the displayed value may not reflect actual anticoagulation — monitor with chromogenic factor X activity instead.

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